

Answer ALL questions. Write your answers in the spaces provided.

Some questions must be answered with a cross ☐. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☐.

- 1** The polysaccharides and proteins that we eat are digested. The products of digestion are then transported in the blood plasma as a solution in water.

(a) Which of the following statements about water are correct?

1. A single molecule of water is composed of an oxygen atom joined to two hydrogen atoms by hydrogen bonds.
2. Water is a solvent.
3. The oxygen in a water molecule has a small negative charge and the hydrogen atoms each have a small positive charge.

(1)

☐ **A** 1 and 2 only

☐ **B** 1 and 3 only

☐ **C** 2 and 3 only

☐ **D** 1, 2 and 3

- (b) The table shows some information about the structure of amylose and amylopectin.

Put **one** cross ☐ in the appropriate box in each row to show whether the structure is present in these molecules.

(3)

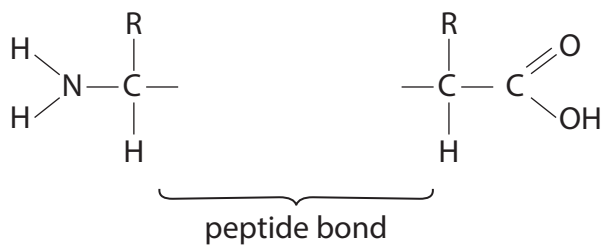
Structure	Structure found in			
	amylose only	amylopectin only	both amylose and amylopectin	neither amylose nor amylopectin
glycosidic bonds	☐	☐	☐	☐
1-4 α bonds	☐	☐	☐	☐
branched side chains	☐	☐	☐	☐



(c) The diagram shows parts of two amino acids.

Complete the diagram to show a peptide bond joining these two amino acids.

(1)



(Total for Question 1 = 5 marks)

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